
RENTASALUD: A WEB-BASED INTERACTIVE ATLAS OF SOCIAL INEQUALITIES AND LIFE EXPECTANCY IN ANDALUSIA (SOUTHERN SPAIN)

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ABSTRACT

Background Interactive tools linking income and life expectancy at fine geographical scales are scarce for southern Europe. **Methods** We linked INE census-section income data (49,888 section-years, 2015–2022) with BDLPA cohort life tables (~637,000 individuals, follow-up 2011–2023) for Andalusia, Spain. Correlation, clustering and GAMs assessed the income–longevity association. **Results** Life expectancy was 79.5 (men) and 84.4 years (women). Eliminating tumours would yield the largest gain (3.93 years, men; 2.83, women). The income–longevity correlation was weak ($r=0.21$, 95%CI 0.59 to 0.81). An R Shiny application integrates all findings in five interactive tabs. **Conclusion** RENTASALUD provides an open tool for exploring health inequalities. Provincial aggregation masks steep census-tract gradients reported nationally. **Keywords:** health inequalities; life expectancy; spatial epidemiology; interactive atlas; Shiny; Andalusia

1 Introduction

Socioeconomic gradients in mortality and life expectancy are extensively documented across Europe (Marmot 2010; Mackenbach et al. 2015). Southern European countries, including Spain, consistently exhibit smaller educational inequalities in mortality than Northern or Eastern European regions, a pattern attributed to the Mediterranean diet, strong social networks and universal healthcare coverage (Mackenbach et al. 2018; Regidor et al. 2012). Despite these smaller relative inequalities, the absolute burden remains substantial: approximately one in five deaths in Spain (roughly 80,000 annually) has been attributed to educational disparities, driven primarily by cardiovascular diseases, smoking-related cancers, respiratory diseases and external causes (Regidor et al. 2014).

At the census-tract level, the evidence on socioeconomic gradients is particularly compelling. Redondo-Sánchez et al. (Redondo-Sánchez et al. 2022) constructed life tables by deprivation quintile across all Spanish census tracts for 2011–2013 and reported that women and men in the most deprived areas lived 3.2 and 3.8 years less, respectively, than those in the least deprived areas. This gradient was more pronounced among men and varied geographically, with higher life expectancy in northern regions and provincial capitals compared to rural areas.

Andalusia, the most populous autonomous community in Spain (approximately 8.5 million inhabitants), has historically exhibited lower incomes and higher unemployment than the national average. Region-specific interactive tools integrating socioeconomic determinants with mortality outcomes remain scarce; most existing platforms aggregate at coarse administrative levels or lack interactive visualisation capabilities.

RENTASALUD addresses this gap by providing an open-source Shiny atlas that combines granular INE census-section socioeconomic data (2015–2022) with life expectancy estimates from the Longitudinal Database of the Andalusian Population (BDLPA, 2011 census cohort, ~637,000 individuals, follow-up through 2023). Our objectives were to: (i) estimate life expectancy by province and sex; (ii) decompose gains by cause of death; (iii) examine the income–longevity association through correlation, clustering and non-linear methods; and (iv) develop an interactive web application accessible to researchers, policymakers and the public.

2 Methods

2.1 Study design and data

We used INE *Atlas de Distribución de Renta de los Hogares* data comprising 49,888 census section–year observations across all eight Andalusian provinces (2015–2022). Variables included median disposable income per consumption unit (modified OECD equivalence scale), mean age, percentage of population under 18 and over 65 years, household size, proportion of single-person households, and population counts.

Life expectancy was estimated from the BDLPA, a 10% representative sample of the 2011 Census of Andalusia (~637,000 individuals) with mortality follow-up to 31 December 2023. The BDLPA provided age at census, sex, province of residence, survival status (alive, emigrated from Andalusia, deceased), and cause of death coded into 10 major groups: circulatory diseases, malignant tumours, endocrine diseases, infectious diseases, respiratory diseases, digestive diseases, nervous system diseases, external causes, alcohol-related causes and remaining causes.

2.2 Life expectancy estimation

We constructed abridged period life tables using Chiang’s method (Chiang 1968) with 5-year age intervals (0–4 to 85–89) and an open-ended 90+ interval. Cause-deleted life tables estimated gains from eliminating each cause. Life expectancy was computed at the provincial level (eight provinces $\times$ two sexes = 16 tables). Census-section estimation was impossible because the BDLPA individual identifier is sequential and contains no geographical information beyond province.

2.3 Income–longevity association

We examined provincial mean income versus life expectancy using Pearson and Spearman correlations, hierarchical clustering (Ward’s linkage on z-scores), and a generalised additive model (GAM) with penalised cubic spline (k=4). Analyses are exploratory given n=8 provinces.

2.4 Ethics

Approved by the University of Granada Ethics Committee (SOCESVIDA, SICEIA-2025-000814, 7 March 2025). Anonymised BDLPA data were provided by the IECA; informed consent was not required.

3 Results

3.1 Life expectancy in Andalusia

Life expectancy at birth was 79.5 years for men and 84.4 years for women, a sex gap of 4.9 years consistent with national figures and reflecting male excess mortality from cardiovascular diseases, lung cancer and external causes. Life expectancy decreased progressively with age for both sexes: at age 65, remaining life expectancy was 19.1 years (men) and 21.9 years (women); at age 85, it was 6.7 and 7.2 years, respectively. The complete period life table by age band is available in the online application.

Provincial variation was modest but consistent across sexes (Table 1). Málaga showed the highest life expectancy (79.6 men, 84.6 women), while Cádiz had the lowest (78.8 and 83.7 years). The provincial range—0.8 years for men and 0.9 years for women—is considerably narrower than the socioeconomic gradient of 3.8 and 3.2 years reported at the census-tract level (Redondo-Sánchez et al. 2022), likely reflecting aggregation bias.

Table 1: Life expectancy at birth by province and sex. Range: 0.8 years (men), 0.9 years (women).

Province	Men (years)	Women (years)
Almería	79.1	84.0
Cádiz	78.8	83.7
Córdoba	79.5	84.5
Granada	79.3	84.2
Huelva	79.0	83.9
Jaén	79.4	84.3
Málaga	79.6	84.6
Sevilla	79.2	84.1

3.2 Cause-of-death decomposition

Malignant tumours were the largest contributor to potential life expectancy gain for men (3.93 years) and circulatory diseases for women (2.89 years). Combined, these two groups account for ~7 years (men) and ~6 years (women). Full results by cause are presented in Table 2.

Table 2: Gains in life expectancy (years) if each cause were eliminated.

Cause	Men	Women
Malignant tumours	3.93	2.83
Circulatory diseases	3.15	2.89
Respiratory diseases	1.13	0.69
External causes	0.67	0.33
Digestive diseases	0.57	0.48
Nervous system diseases	0.50	0.57
Infectious diseases	0.41	0.35
Endocrine diseases	0.28	0.32
Alcohol-related causes	0.07	0.02
Remaining causes	1.13	1.19

3.3 Income–life expectancy association

The Pearson correlation between provincial mean income and life expectancy was weak and non-significant ($r = 0.21$; 95% CI: 0.59 to 0.81; $p = 0.62$). Spearman’s ρ confirmed this ($\rho = 0.07$, $p = 0.87$). Cluster analysis identified two

groups: Cluster 1 (Málaga, Sevilla, Córdoba, Granada; income 13,341–14,338; life expectancy 81.75–82.10 years) and Cluster 2 (Almería, Cádiz, Huelva, Jaén; 12,699–13,172; 81.25–81.85 years). The GAM showed no evidence of non-linearity (EDF = 1.32, $p = 0.48$).

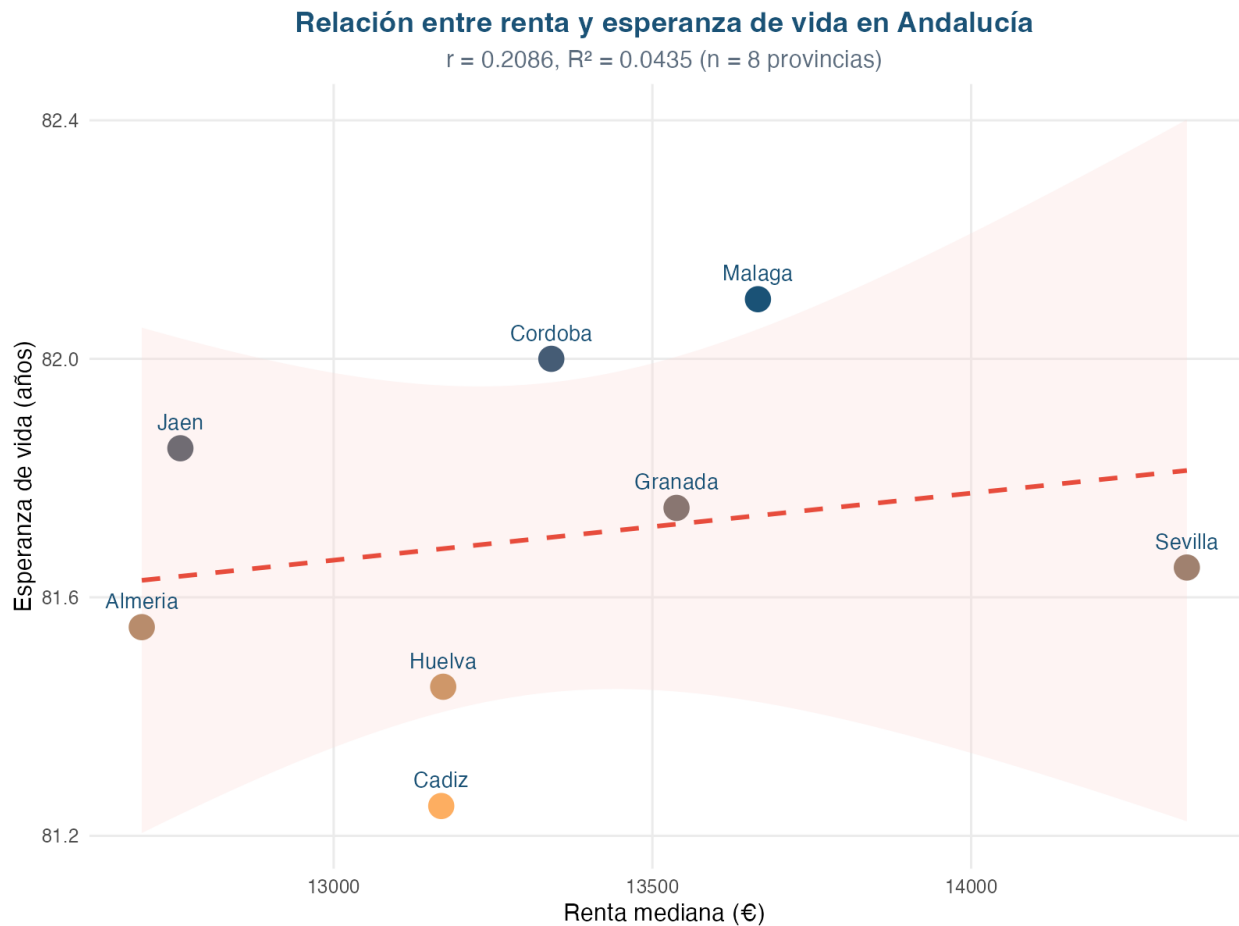


Figure 1: Scatter plot with loess smoother and linear regression ($R^2 = 0.04$, $p = 0.62$).

3.4 Interactive application

The RENTASALUD Shiny application integrates all findings across five interactive tabs. The *Explorar* tab features choropleth maps with a year slider (2015–2022) for animating temporal changes across 12 indicators. The *Asistente IA* tab enables natural-language queries via large language models (Ollama, OpenAI or Gemini). The *Relaciones* tab presents scatter plots with loess smoothers, cluster dendrograms and grouped bar charts of life expectancy by province and sex. The *Causas Mortalidad* tab provides interactive cause-of-death decomposition with horizontal bar charts, searchable tables and age-band life expectancy curves. The *Metodología* tab documents all indicator definitions. Source code: github.com/migariane/RENTASALUD.

4 Discussion

4.1 Principal findings

This study provides the first integrated, interactive atlas of income, demography and life expectancy for Andalusia. Life expectancy at birth (79.5 years for men, 84.4 for women) falls below the Spanish national average (~80.5 and ~86.0 years, respectively) (Instituto Nacional de Estadística 2024), consistent with Andalusia's historically lower socioeconomic indicators. The dominance of malignant tumours and circulatory diseases as causes of premature mortality—together accounting for approximately 7 years of potential life expectancy gain in men and 6 in women—aligns with European preventable mortality patterns (Beltrán-Sánchez et al. 2008; Mackenbach et al. 2015) and reinforces the continued importance of cancer screening, cardiovascular risk prevention and smoking cessation interventions. The larger tumour

impact in men (+1.1 years vs women) reflects Spain's historically higher male smoking prevalence, a gap that is narrowing in younger cohorts.

4.2 Geographical scale and the income–longevity gradient

The weak and non-significant provincial-level income–longevity correlation ($r = 0.21$; 95% CI: -0.59 to 0.81) contrasts sharply with the steep census-tract gradients reported by Redondo-Sánchez et al. (Redondo-Sánchez et al. 2022), who found a gap of 3.8 years for men and 3.2 years for women between the most and least deprived census tracts nationally. This discrepancy, spanning an order of magnitude, illustrates the modifiable areal unit problem (MAUP): provincial aggregation masks substantial intra-provincial heterogeneity in both income and mortality. The Southern European pattern of smaller health inequalities (Mackenbach et al. 2018; Regidor et al. 2012) does not imply absence of a socioeconomic gradient; rather, the gradient operates at finer spatial scales and may be mediated by different mechanisms—including dietary patterns, social networks and universal healthcare—compared to Northern or Eastern Europe. These findings imply that area-based interventions should target the most deprived census sections, and RENTASALUD is designed precisely to facilitate this through interactive sub-provincial mapping.

4.3 Strengths and limitations

The main strength is the integration of individual-level longitudinal mortality follow-up from the BDLPA (~637,000 individuals, 12 years) with granular census-section socioeconomic data within an open-source, reproducible platform featuring an AI-powered query interface. Several limitations should be acknowledged. First, life expectancy was estimated only at the provincial level, the most important limitation; future work should pursue BDLPA–Padrón Continuo linkage to recover census-section codes for small-area Bayesian estimation. Second, period life tables assume constant age-specific mortality rates, which may not hold during periods of rapid change. Third, sample sizes limit precision at extreme ages (0–4 and 90+) when disaggregating by province and sex. Fourth, there is a partial temporal mismatch between income data (2015–2022) and mortality follow-up (2011–2023). Fifth, cause-deleted decomposition assumes independence between causes, meaning gains from eliminating multiple causes are not additive.

5 What is already known on this topic

- Socioeconomic deprivation is strongly associated with mortality in Spain, with a gap of 3.2–3.8 years between most and least deprived census tracts.
- Southern Europe exhibits smaller educational inequalities in mortality than Northern or Eastern Europe.
- Interactive tools integrating income and life expectancy at fine geographical scales remain scarce.

6 What this study adds

- RENTASALUD provides the first open-source interactive atlas of census-section income and life expectancy for Andalusia.
- Provincial-level income–longevity correlation is weak ($r = 0.21$), in contrast to steep census-tract gradients.
- Malignant tumours and circulatory diseases account for ~7 years of potential life expectancy gain in men.
- An AI-powered natural language interface enables non-specialist querying of the data.

7 Acknowledgements

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8 Data Availability

Source code: github.com/migariane/RENTASALUD. Shiny app: [shinyapps.io link]. BDLPA microdata available from IECA on request.

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